

```
main.cpp  Run  Clear
1 #include<iostream>
2 using namespace std;
3 int main ()
4 {int id;
5   float cgpa;
6   string name;
7   cout << "Enter id, name, cgpa :";
8   cin >> id >> name >> cgpa;
9   cout << "Name : " << name << endl;
10  cout << "Id : " << id << endl;
11  cout << "Cgpa : " << cgpa << endl;
12  cin.ignore();
13  cout << "Enter your full name :";
14  string fullname;
15  getline(cin, fullname);
16  cout << "fullname : " << fullname << endl;
17  return 0;
18 }
19
```

Output

```
Enter id, name, cgpa :84 etu 3.89
Name : etu
Id : 84
Cgpa : 3.89
Enter your full name :anika tasnim etu
fullname : anika tasnim etu

=== Code Execution Successful ===
```

```
main.cpp  Run  Clear
9 void info ()
10 {
11   cout<<"name :"<<name<<endl;
12   cout<<"id:"<<id<<endl;
13   cout<<"Cgpa : "<<cgpa <<endl;
14 }
15 };
16 int main ()
17 {
18   student s1,s2;
19   s1.name="etu";
20   s1.id=1;
21   s1.cgpa=3.88;
22   s1.info();
23   s2.name="Tasnim";
24   s2.id=2;
25   s2.cgpa=3.78;
26   s2.info();
27   return 0;
28 }
29
```

Output

```
name :etu
id:1
Cgpa :3.88
name :Tasnim
id:2
Cgpa :3.78

=== Code Execution Successful ===
```

```
main.cpp  Run  Clear
16
17 void input(float discount)
18 {
19   float discountprice, newprice;
20   discountprice=price*discount/100;
21   newprice=price-discountprice;
22   cout<<"Book details:"<<"\n"<<"title:"<<title<<"\n"<<"price:"
    <<price<<"\n"<<"discount"<<newprice<<endl;
23 }
24 };
25 };
26 int main ()
27 {
28   int discount;
29   Book b1;
30   b1.info();
31   cout<<"Enter discount:"<<endl;
32   cin>>discount;
33   b1.input(discount);
34   return 0;
35 }
```

Output

```
Enter book title :
Untold story
Enter book author :
Anika tasnim
price:
9998.88
Enter discount:
34
Book details:
title:Untold story
price:9998.88
discount6599.26

=== Code Execution Successful ===
```

```
main.cpp  Run  Clear
4
5 int id; //if not declared then private
6 public :
7 //setter
8 void setid(int i){
9   id=i;
10 }
11
12 //getter
13 int getid()
14 {
15   return id;
16 }
17 };
18 };
19 int main ()
20 {
21   student s1;
22   s1.setid(84);
23   cout<<"Student id:"<<s1.getid(); //gets value
24   return 0;
25 }
```

Output

```
Student id:84

=== Code Execution Successful ===
```

```
main.cpp  Run  Output  Clear
11
12     age=a;
13     else cout<<"invalid age "<<endl;
14     name=n;
15 }
16 int getage()
17 {
18     return age;
19 }
20 }
21
22 string getname(){
23     return name;
24 }
25
26 };
27 int main (){
28
29     person p1;
30     p1.setdata("Etu",22);
31     cout<<"name:"<<p1.getname()<<" " <<"age"<<p1.getage()<<endl;
```

```
name:Etu  age22
--- Code Execution Successful ---
```

```
main.cpp  Run  Output  Clear
8 void show();
9 ~myclass(); //destructor
10
11 };
12 myclass ::myclass ()
13 {
14     cout<<"constructor"<<endl;
15     a=100;
16 }
17 }
18 void myclass ::show(){
19     cout<<"value of a:"<<a<<endl;
20 }
21 ~myclass::~myclass (){
22     cout<<"destructor"<<endl;
23 }
24 int main(){
25     myclass m1,m2;
26     m1.show();
27     m2.show();
28     return 0;
```

```
constructor
constructor
value of a:100
value of a:100
destructor
destructor
--- Code Execution Successful ---
```

```
main.cpp  Run  Output  Clear
12
13 student::student(int i, string n) {
14     id = i;
15     name = n;
16 }
17
18 void student::show() {
19     cout << "id:" << id << " " << "name:" << name << endl;
20 }
21
22 student::~student() {
23 }
24
25 int main() {
26     student s1(84, "etu");
27     student s2(83, "paro");
28     s1.show();
29     s2.show();
30     return 0;
31 }
32
```

```
id:84  name:etu
id:83  name:paro
--- Code Execution Successful ---
```

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main.cpp

```
34 }
35
36 void student::display() {
37     cout << "name:" << name << "   age:" << age << "   cgpa:" << cgpa
38     << endl;
39 }
40 student::~student() {
41     cout << "destructor" << endl;
42 }
43
44 int main() {
45     student s1;
46     student s2(" ");
47     student s3("N/A", 20, 3.5);
48     s1.display();
49     s2.display();
50     s3.display();
51     return 0;
52 }
53 }
```

Output

```
constructor
enter name:
anika tasnim
name:N/A age:0 cgpa:0
name:anika tasnim age:0 cgpa:0
name:N/A age:20 cgpa:3.5
destructor
destructor
destructor

=== Code Execution Successful ===
```

main.cpp

```
30     length = 1;
31     width = w;
32 }
33
34 void rectangle::display() {
35     cout << "length:" << length << "   width:" <<
36     width << endl;
37 }
38 rectangle::~rectangle() {
39 }
40
41 int main() {
42     rectangle r1;
43     r1.display();
44 }
```

Output

```
length:1 width:1
length:2 width:3
invalid
length:1 width:1

=== Code Execution Successful ===
```

main.cpp

```
24 public:
25     student(string n, int a, int i, float c) : person(n, a) {
26         id = i;
27         cgpa = c;
28         cout << "student" << endl;
29     }
30     void displaystudent() {
31         displayperson();
32         cout << "id:" << id << " cgpa:" << cgpa << endl;
33     }
34 };
35
36 int main() {
37     student s1("etu", 22, 84, 3.96);
38     s1.displaystudent();
39     return 0;
40 }
41 }
```

Output

```
person
student
name:etu age:22
id:84 cgpa:3.96

=== Code Execution Successful ===
```

main.cpp

Share

Run

19

string brand;

20

public:

21

car(int s, string b) : vehicle(s) {

22

brand = b;

23

cout << "car is called" << endl;

24

}

25

void display() {

26

displayspeed();

27

cout << "brand:" << brand << endl;

28

}

29

};

30

31

int main() {

32

car c(22, "toyota");

33

c.display();

34

return 0;

35

}

36

Output

Clear

vehicle is called

car is called

speed:22

brand:toyota

=== Code Execution Successful ===

main.cpp

Share

Run

25

};

26

27

class D : public B, public C {

28

public:

29

D() : A(23) {

30

cout << "D class" << endl;

31

}

32

void show() {

33

cout << "x:" << x << endl;

34

}

35

};

36

37

int main() {

38

D obj;

39

obj.show();

40

return 0;

41

}

42

Output

Clear

A class

B class

C class

D class

x:23

=== Code Execution Successful ===

main.cpp

Share

Run

36

person(n, a), teacher(n, a, eid), researcher(n, a,

area) {

37

cout << "professor" << endl;

38

}

39

void show() {

40

cout << "name:" << name << " age:" << age

41

<< " employee:" << employeeid

42

<< " research area:" << researcharea << endl;

43

}

44

};

45

46

int main() {

47

professor p("DR. Anika", 22, 1084, "Artificial

Intelligence");

48

p.show();

49

return 0;

50

}

51

Output

Clear

person

teacher

researcher

professor

name:DR. Anika age:22 employee:1084 research area:Artificial

Intelligence

=== Code Execution Successful ===